

Retrieval, document review, and grounding

This is the *read path*. Briefing 3 wrote `document_pages`, `document_outline_spans`, and `document_chunks` (pgvector + English FTS). This briefing queries those rows. Search finds candidate pages. Document review walks a queued set. Grounding decides whether a drafted fact may persist.

Reads what ingest wrote. There is no Pinecone, no Elasticsearch, no BM25 index. Vector = `document_chunks.embedding <=> (HNSW cosine)`. Keyword = `to_tsvector('english', contextual_text) @@ websearch_to_tsquery`. Both arms fuse with reciprocal rank fusion (k=60). See briefing 3 if those tables are a blank.

1. What search reads (from briefing 3)

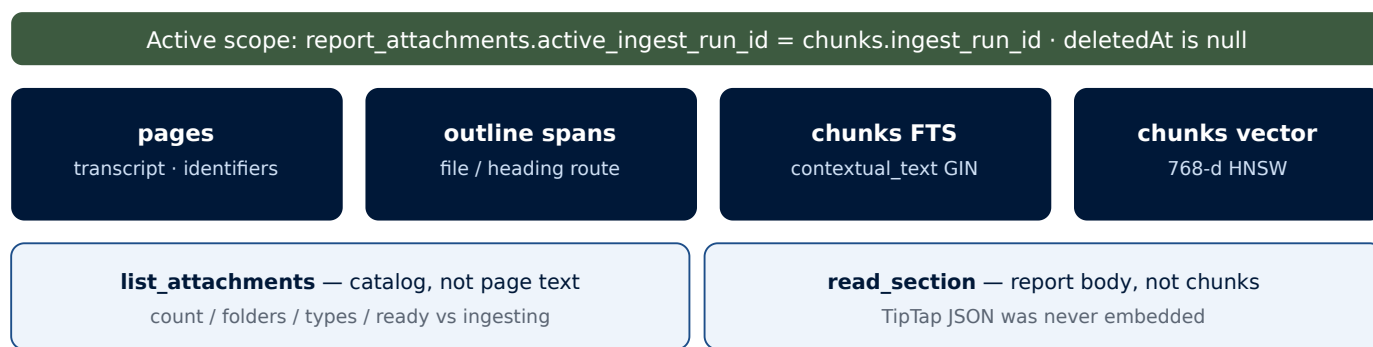


Figure 1. File-set questions use `list_attachments`. Which files mention a fact inside a PDF use `search_documents`. The report body uses `read_section`.

2. Hybrid search pipeline

`searchReportDocuments / searchReportDocumentsDetailed` in `src/lib/attachments/retrieval.ts`. Document chat uses hybrid. Analytics chat is keyword-first at the tool layer (same function, mode: "keyword" — still Postgres FTS, still not BM25).

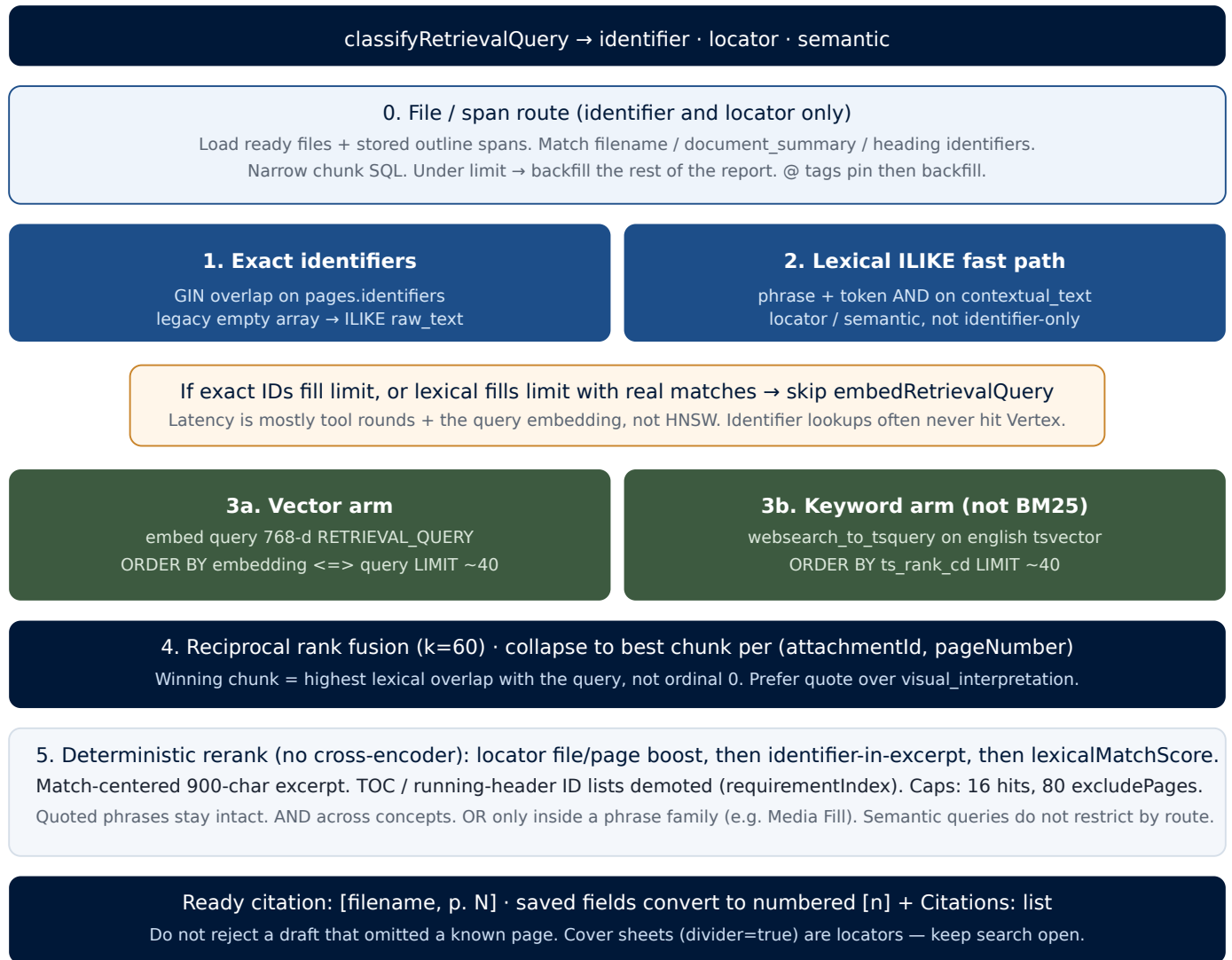


Figure 2. Two SQL lists, fused in application code. Adding Elasticsearch would duplicate this and leave the harness still deciding when to grep.

Query kind	What runs
identifier	Requirement-like IDs from requirementIds(). Exact SQL first. If those hits fill limit, skip embed. Prepend remaining hybrid RRF.
locator	Page / filename wording. Route first. Lexical fast path. Then hybrid if needed.
semantic	No extra identifier SQL. Vector + FTS + RRF. Unrestricted semantic stays two SQL calls (vector + keyword).

3. Search loop (when the harness hides grep)

searchLoopDirective in search-loop.ts. Shared by Document and Analytics, with different “this was a real cited page” rules.

Event this turn	search_documents	ask_user (Document)
Cited hit, locate/read, or two empty greps	Hide (directive read)	Hide until a page is actually read
Every hit is a cover sheet (dividerHits)	Stay open	Hide until read_document_page
Tool returned keepSearchOpen or unsupported_facts	Stay open	—
Active document review	Stay open (review tools own the step)	—
Analytics: a real cited page	Hide for the rest of the turn	Never ask which page — search/scan and say found or not

Never quiz the page number. After any grep, Document chat hides ask_user until read_document_page. Outline locates; it does not unlock a quiz.

4. Document review state machine

Empty inventories and open-set drafts with distributed evidence use a page walk, not another grep. Tools: `start_document_review` → `continue_document_review` → `finish_document_review`. Phase lives on the in-turn `DocumentReviewSession`.



start `planReviewPages` queues objective-matching pages (floor 8). Nearby same-file pages only — no leftover pad to 2500. `REVIEW_PAGE_FETCH_CAP = 2500` is listing safety so one file cannot starve others. It is not a walk cap.

continue 8-wide extract pool in one call. Stop starting new batches after ~60s (`REVIEW_CONTINUE_BUDGET_MS`). Leave 20s of the 270s abort window for persist. Transcript-backed batches skip the extract LLM.

finish Returns a capped findings sample (60). Seeds the ledger with `reviewedEvidence` (every walked page). Rehydrate only `status=complete` and `truncated != true`. `coverageKey = selected documents' page counts + |obj|`.

already_complete Same `|obj|`: just finished → do not reset into another walk. Fill with `edit_table`.

Figure 3. `prepareDocumentReviewStep` forces the tool for the current phase. A leftover qualification finish does not unlock Calibration.

Coverage identity

Piece	Why it exists
<code>coverageKey</code> includes <code> obj </code> :	A finished calibration walk must not satisfy monitoring / associated instruments.
<code>coverageKeySatisfiesObjective</code>	Rehydrate when attachments are unchanged <i>and</i> the objective still matches.
Floor-8 skip of the PRQR	Does not unlock <code>edit_table</code> . CSV-OQ headers while the evidence file was never queued is not a complete inventory.
Matching finish this turn	Hide review tools even if the table is still empty — that emptiness is why you reviewed. Next step is <code>edit_table</code> .
Explicit retrieval pushback	Skips rehydrate. Queued inventory keeps comprehensive (<code>planKeepsComprehensive</code>).

placeholder_fill ≠ page walk. A populated table that still has `<date>` / `<identifier>` is adaptive closed-set search (`placeholder-fill.ts`). Header-only seeded tables (blank data cells) are empty inventories and still need a matching finished review.

5. Citation ledger and fact gate

Hard facts (dates, identifiers, measured numbers) in `draft_field` / `propose_edit` / `edit_table` must match a quote on a page this turn retrieved. Empty ledger or pages with no served quotes **fail open**.



groundDraftText rewrite invented p. N · match each fact to a served quote · park citations

demo / Convergent · flag

Persist the draft. Attach Traceability on the suggestion card. Audit `claim_verified` / `claim_unsupported`.

MJ · block

Do not persist. Return `unsupported_facts`.
One closed-set repair search, re-ground.
No LLM subagent.

MJ also refuses leftover `<date>` / `<identifier>` / `<number>` until a same-turn page read or non-divider search.

If repair adds no new quoted page, placeholders may persist — do not invent the value. Analytics `write_column` is not gated.

`finish_document_review` seeds the ledger with `reviewedEvidence` so later drafts in the turn can hydrate those transcripts.

Follow-up turns keep a slim `citationDigest` ([filename, p. N] + short summary), not the full 273-page findings array.

Figure 4. The model can still hallucinate a batch number. The harness is the last gate before an `ai_fix` comment is written.

What counts as a quote

- Search hit snippet (budget 900 chars)
- `read_document_page` transcript (12k chars)
- Every walked page on finish, not the 60-finding sample

What does not count

- The Documents index / `documentSummary`
- Filename-only [`file.pdf`] with no page
- Cover-sheet hits (`divider=true`)
- Prior-turn findings unless compacted into `citationDigest`

6. Compaction, budgets, abort

Every model step resends system prompt + tool JSON schemas + the growing tool-result transcript. A 273-page finish that keeps full findings will 500 the next message. That is why compaction exists.

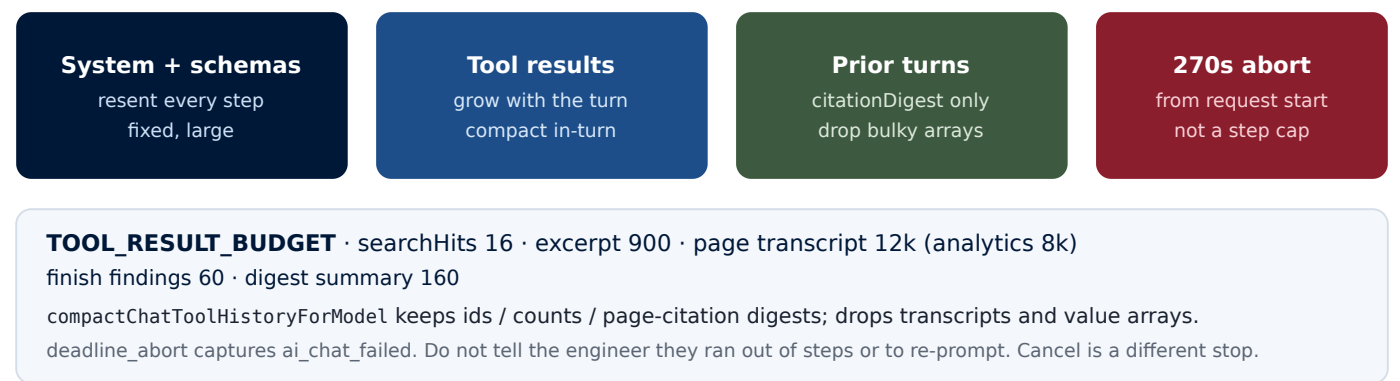


Figure 5. There is no per-turn tool-step cap. Loop guards live in prepareStep. Hobby function max duration must stay above 270s.

Constant	Value	File
CHAT_SERVER_ABORT_MS	270_000	assistant-turn.ts
REVIEW_CONTINUE_BUDGET_MS	60_000	document-review.ts
REVIEW_CONTINUE_DEADLINE_MARGIN_MS	20_000	leave room to persist after continue
REVIEW_EXTRACT_CONCURRENCY	8	in-flight extracts in one continue
REVIEW_OBJECTIVE_PAGE_FLOOR	8	review-page-plan.ts
REVIEW_PAGE_FETCH_CAP	2500	listing safety, not a walk cap

Debug a “it didn’t search” bug in this order. (1) Latest-turn intent — was it social? (2) ChatTurnPlan.retrievalPolicy — focused skips grep. (3) prepareReportChatStep activeTools on that step — search already hidden? (4) Langfuse generation — did the model see the tool? (5) Retrieval eval (pnpm retrieval-eval) if the page was wrong, not if the tool never ran.

7. Clicking a citation

In the document, chat, or suggestion Sources, [filename, p. N] (or numbered [n] mapped to a parked source) opens or reuses that attachment’s canvas tab and scrolls the PDF. Combined [file A, p. N, file B, p. M] splits into one link per file. Word preview cannot jump to a page. Missing or ambiguous filenames toast. Open canvases stay mounted while you switch to Report so the PDF does not re-parse.

Andrei · retrieval · briefing 3 wrote the tables this PDF queries · code is source of truth